

Same but different: The incremental interpretation of embedded topic-marked subjects in Korean

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33rd Japanese/Korean
Conference

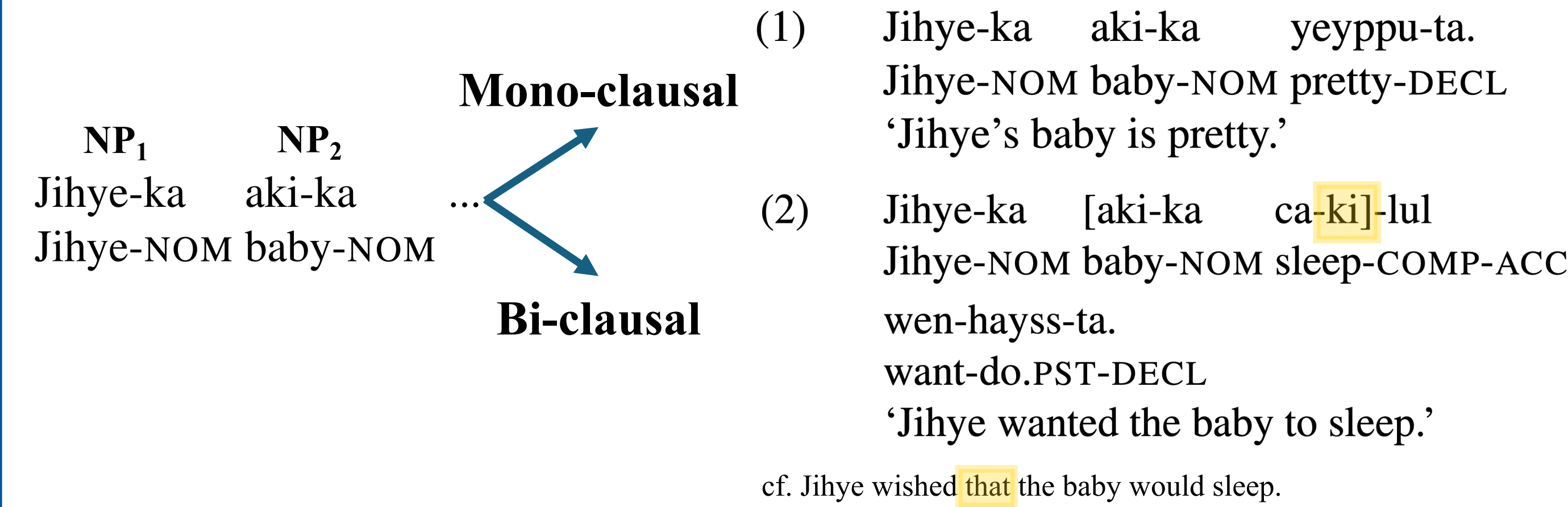


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1 How does the parser know if the structure is mono- or bi-clausal?

- Korean** allows the sequence of **two arguments** with the **same case marker** in a sentence.



- Unlike English, **Korean complementizers** are attached **after** the embedded predicate.
- When** does the parser **commit** to a structure by incremental processing?

- 1-a**
- Case markers serve as **cues to mark clause boundaries** (Kim 1999, Miyamoto 2002, and many others).
 - Can an **information-structural topic marker** function the **same** way as case markers do?

2 Topic alternates with nominative case.

- Propositional meaning between (1)-(3) and (2)-(4) are **identical** when NP₁ is marked with the **topic** marker (Huang 1984).
- (3) Jihye-**nun** aki-ka yeppu-ta. (4) Jihye-**nun** [aki-ka ca-ki]-lul
Jihye-**TOP** baby-NOM pretty-DECL Jihye-**TOP** baby-NOM sleep-COMP-ACC
'Jihye's baby is pretty.' wen-hayss-ta.
want-do.PST-DECL
'Jihye wanted the baby to sleep.'
- However, there is a **difference** in information structure (Kuno 1973, Lee & Shimojo 2016);
 - Nominative case marker = discourse-new information, • NP₁ **topic** = **Sentence topic**
Topic marker = discourse-old information

2-a Topic alternates with accusative case.

- Propositional meaning between (1)-(5) and (2)-(6) are **different** when NP₂ is marked with the **topic** marker.
- (5) Jihye-ka aki-**nun** yeppu-ta.
Jihye-NOM baby-**TOP** pretty-DECL
'Jihye's baby is pretty, but not anything else of Jihye.'
- (6) Jihye-ka [aki-**nun** ca-ki]-lul wen-hayss-ta.
Jihye-NOM baby-**TOP** sleep-COMP-ACC want-do.PST-DECL
'Jihye wanted the baby to sleep, but no one else.'
- Topic** marking is interpreted **contrastively** when preceded by another argument (Disregarding focus or intonation) (Lee 2017). • NP₂ **topic** = **Contrastive topic**

2-b Multiple topics.

- Multiple topics are acceptable in Korean, as in Japanese, Italian, German, etc. (Kuno 1973, Rizzi 1997).
- (7) a. Jihye-nun aki-nun coh-ta.
Jihye-TOP baby-TOP like-DECL
'Jihye likes the baby, but does not like anything else.'
- b. Jihye-nun [aki-nun ca-ki]-lul wen-hayss-ta.
Jihye-TOP baby-NOM sleep-COMP-ACC want-do.PST-DECL
'Jihye wanted the baby to sleep, but not anyone else.'
- Topic** marker is **iterative**, but there can only be **one sentence topic** per sentence (NP₁) (Krifka & Musan 2012).
 - Every **other** topic in the sentence is interpreted as **contrastive** (NP₂) (Kuroda 1988).

2-c Embedded topic.

- Topic **cannot be easily embedded**, since the TopP is placed higher than the speech act operator (Tomioka 2016).
- (8) a. Jihye-nun [aki-nun can-ta]-ko mal-hayss-ta.
Jihye-top baby-TOP sleep-DECL-COMP say-do.PST-DECL
'Jihye said that the baby is asleep.'
- b. Jihye-nun mal-hayss-ta [aki-nun can-ta]-ko.
Jihye-top say-do.PST-DECL baby-TOP sleep-DECL-COMP
'Jihye said that the baby is asleep.'
- Root-like** embedded clauses permit sentence topics, such as **complements of verbs of communication** (e.g., say, report) (Hooper & Thompson 1973);
 - Jihye* = **sentence topic** of the **matrix** clause,
 - aki* (baby) = **sentence** topic of the **embedded** clause

3 Research questions.

- Can topic marker guide incremental clause interpretation (like the nominative case marker)?
- Does the parser commit to an interpretation *before* or *at* the complementizer *-ko*?

3-a Experiment design – Self-paced reading.

- A **2 x 2 factorial** experimental design via **moving-window self-paced reading (non-cumulative)** method in PCIBex Farm (Zehr & Schwarz 2018).
- Matrix** subject marking (nominative vs. topic) x **Embedded** subject marking (nominative vs. topic) [independent variables]
- Reading time** [dependent variable]
- 16 stimuli sets and 32 filler sentences; • Pseudo-randomized order;
- 8 random comprehension questions; • Four item sets made with Latin-square design

3-b Predictions.

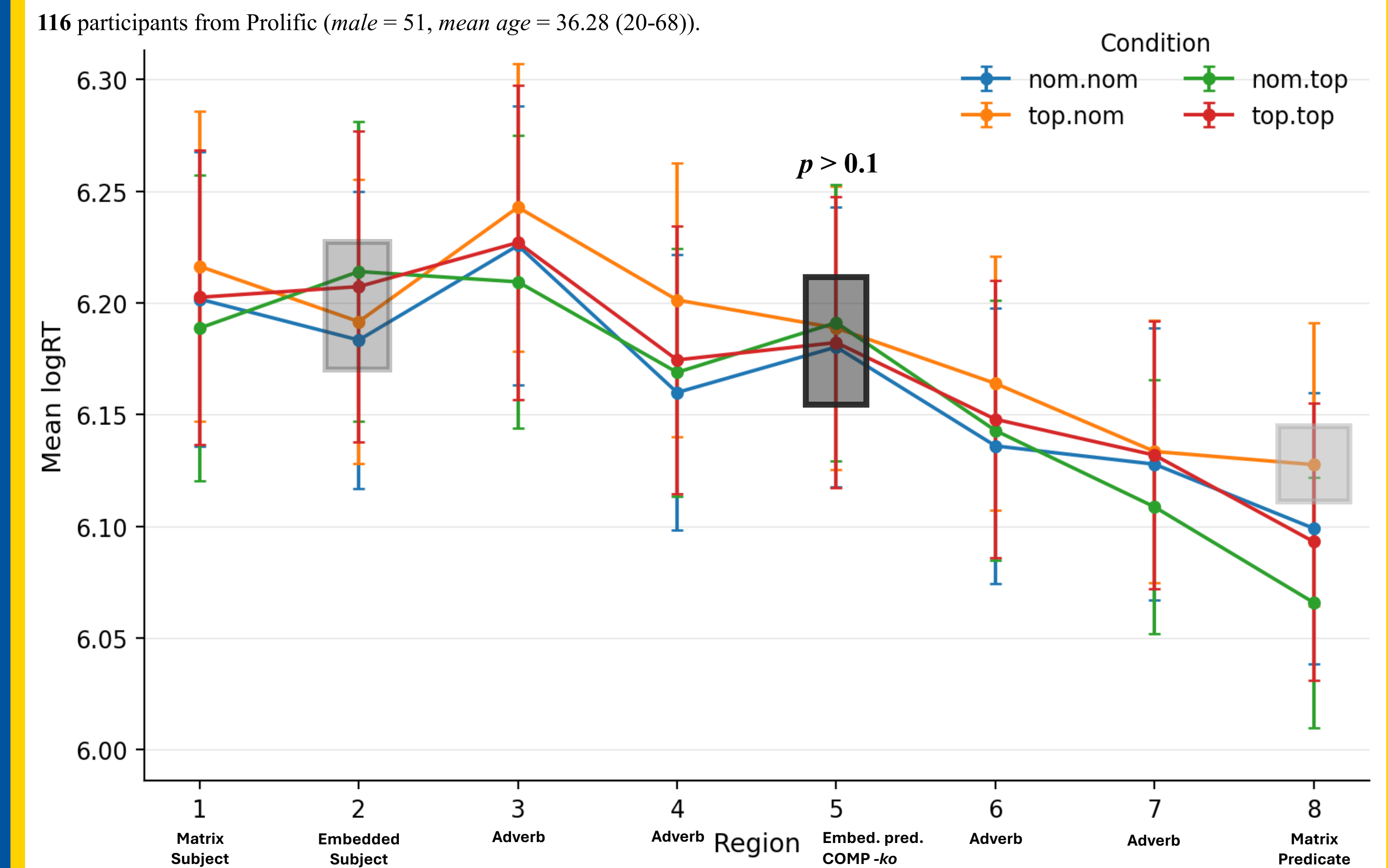
- Garden path model** (Frazier & Fodor 1978, Just & Carpenter 1980, Frazier & Clifton 1996): Minimal attachment assumes mono-clause; late closure at embedded predicate = longer reading time
- Frequency model** (Trueswell et al. 1994): Processing difficulty (by frequency in the corpus (Cho & Tollan 2026)):
NOM-NOM > TOP-TOP > NOM-TOP > TOP-NOM

3-c Materials.

		Embedded Subject					
		NOM			TOP		
Matrix Subject	NOM	John-i₁ Kate-ka₂ kongwen-eyse₃ John-NOM Kate-NOM park-at cacwu₄ chwumchw-ess-ta-ko₅ ecey₆ often dance-PAST-DECL-COMP yesterday na-eykey₇ malh-ayss-ta₈. me-to say-PST-DECL 'John said to me yesterday that Kate often dances at the park.'	John-i₁ Kate-nun₂ kongwen-eyse₃ John-NOM Kate-TOP park-at cacwu₄ chwumchw-ess-ta-ko₅ ecey₆ often dance-PAST-DECL-COMP yesterday na-eyke-y₇ malh-ayss-ta₈. me-to say-PST-DECL 'John said to me yesterday that Kate often dances at the park.'				
	TOP	John-un₁ Kate-ka₂ kongwen-eyse₃ John-TOP Kate-NOM park-at cacwu₄ chwumchw-ess-ta-ko₅ ecey₆ often dance-PAST-DECL-COMP yesterday na-eykey₇ malh-ayss-ta₈. me-to say-PST-DECL 'John said to me yesterday that Kate often dances at the park.'	John-un₁ Kate-nun₂ kongwen-eyse₃ John-TOP Kate-TOP park-at cacwu₄ chwumchw-ess-ta-ko₅ ecey₆ often dance-PAST-DECL-COMP yesterday na-eykey₇ malh-ayss-ta₈. me-to say-PST-DECL 'John said to me yesterday that Kate often dances at the park.'				

* The number after each word refers to the region of the sentence.

4 Parser commits to an interpretation before *-ko*.



4-a Statistical analysis.

Linear mixed-effects regression analysis (region-by-region)

```
model <- lmer(
  logRT ~ matrix_c * embed_c +
    (1 + matrix_c * embed_c | participant) +
    (1 + matrix_c * embed_c | number),
  data = data
)
```

Fitted to the parsimonious model (Matuschek et al. 2017)

	Effect	Estimate (β)	SE	t	p
Region 2 (Embedded subject)	Intercept	6.249	0.074	84.23	<.001
	Matrix	-0.027	0.018	-1.50	.13
	Embedded	0.047	0.018	2.57	** .01
	Interaction	-0.002	0.036	-0.07	.95

Effect: β < 0: **nominative** has bigger effect on RT
β > 0: **topic** has bigger effect on RT

	Effect	Estimate (β)	SE	t	p
Region 8 (Matrix predicate)	Intercept	6.099	0.0476	128.16	<.001
	Matrix	0.027	0.011	2.39	*.017
	Embedded	-0.027	0.116	-2.32	*.038
	Interaction	0.0325	0.022	1.48	.143

- NOM-TOP, TOP-TOP** conditions are read **slower** in the embedded subject (Region 2).
- Possible sources** for slowdown at Region 2:
 - **Information structural integration**
 - Reduced preference for NP2 topic
 - Early clause-structure prediction
- Wrap-up effect** (Just & Carpenter 1980) is observed in the **TOP-NOM** condition.

5 Take away points.

- Topic marker serves as an **early information-structural cue** during incremental sentence processing.
- Embedded topic marker increased processing cost** at Region 2.
- There is **insufficient** evidence to conclude that reanalysis was observed at *-ko* (Region 5).

Selective reference.

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